IMMERSIONS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3669-3682, following the statement at 3661-3667.

Editorial note. The identification X_{S'} cong Y_{S'} x_Y X is read off the displayed 3x3 diagram (diagram-anchored reading, same precedent as cards L26341/L26383); the three immersion variants all reduce to the named engine.

Allowed standard prerequisites (editorial).

- Immersions stable under fibre products with a fixed factor. lemma-fibre-product-immersion (card L26413, cited at ~3681). Concluding engine.

Source reference. Original lemma 01JU: sources/stacks-schemes/pinned-original-source.tex, source lines 3547–3547. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an immersion (resp. closed immersion, resp. open immersion) of schemes over S. Then any base change of f is an immersion (resp. closed immersion, resp. open immersion).*

Proof. We can think of the base change of f via the morphism $S' \to S$ as the top left vertical arrow in the following commutative diagram:

$$\begin{array}{ccc} X_{S'} & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y_{S'} & \longrightarrow & Y \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

The diagram implies $X_{S'} \cong Y_{S'} \times_Y X$, and the lemma follows from Lemma 01JU. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.